**Name : Shivani Anand,Shravani Kaitikwar**
**Batch : B4_b2**
**Roll no. : 21,22**
**SEPM**

## Prac-7

**AIM: cocomo cost estimation of three types specific:**
**Organic**
**Semi-detached**
**Embedded**
**of our topic program any language**

**Project title: Student Attendance Management System**

**Python Code:**

```
# cocomo cost estimation
# project: student attendance management system

# estimated lines of code for the project
loc = 3000

# convert loc into kloc
kloc = loc / 1000

# monthly cost of one developer
cost_per_person_month = 50000


# function to calculate cocomo
def cocomo(mode, a, b, c, d):

    effort = a * (kloc ** b)
    development_time = c * (effort ** d)
    average_staff = effort / development_time
    total_cost = effort * cost_per_person_month
```

```
    print("\n")
    print(mode)
    print("KLOC:", kloc)
    print("Effort:", round(effort, 2), "person-months")
    print("Development Time:", round(development_time, 2), "months")
    print("Average Staff:", round(average_staff, 2), "persons")
    print("Total Cost: ₹", round(total_cost, 2))


# organic model
cocomo("ORGANIC", 2.4, 1.05, 2.5, 0.38)

# semi-detached model
cocomo("SEMI-DETACHED", 3.0, 1.12, 2.5, 0.35)

# embedded model
cocomo("EMBEDDED", 3.6, 1.20, 2.5, 0.32)
```

Output:

```
ORGANIC
KLOC: 3.0
Effort: 7.61 person-months
Development Time: 5.4 months
Average Staff: 1.41 persons
Total Cost: ₹ 380328.23


SEMI-DETACHED
KLOC: 3.0
Effort: 10.27 person-months
Development Time: 5.65 months
Average Staff: 1.82 persons
Total Cost: ₹ 513413.24


EMBEDDED
KLOC: 3.0
Effort: 13.45 person-months
Development Time: 5.74 months
Average Staff: 2.34 persons
Total Cost: ₹ 672694.71
```

```
Total Cost: ₹ 672694.71
```

[3]:
```
loc = int(input("Enter estimated lines of code: "))
kloc = loc / 1000

cost_per_person_month = float(
    input("Enter cost per person-month: ₹")
)
```

```
Enter estimated lines of code:  50
Enter cost per person-month: ₹ 500
```

[ ]: